Summary and Interpretation(customer purchasing behavior)

Key Descriptive Statistics

Column	Mean	Median	Mode	IQR	VAR	STD	RANGE
TOTAL_BILL	19.78	17.80	13.42	10.77	79.25	8.90	47.74
TIP	2.99	2.9	2	1.565	1.91	1.38	9
SIZE	2.56	2	2				

Variation and spread insight

- Standard deviation(8.90)
  - It means that most of the bill is above 9$ or below 9$ (high variation)
- Range(47.74)
  - it means customer base in restaurant is belong to every type of budget(from low to high order).
- IQR(10.77)
  - difference between 50% customer spending is around 11$ it means there is no much unpredictability in their spending habit.

Central Tendency Insight

- Mean>median>mode-It means shape of total bill is rightly skewed which tells that there are some outlier which is stretching average spending of total bill.

- Common spending level 13.42$ (mode) most frequent amount-most significant portion of customer base is looking for budget friendly or value meals.

TIPPING BEHAVIOUR

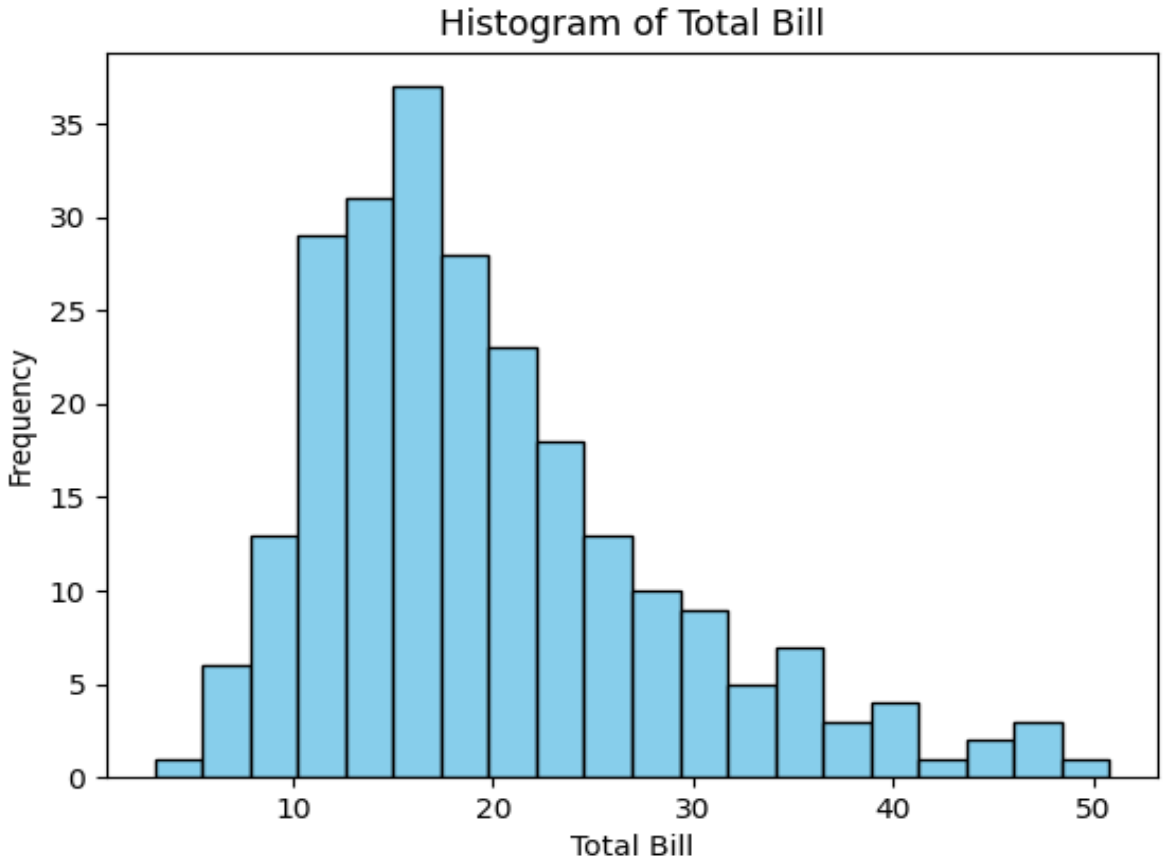
- Predictable Tipping Behaviour-as there is no much difference between median and avg tip
- mode of tip is 2.0 most frequent tip given(indicate flat tip culture among customer rather percentage based approach)

GROUP DYNAMICS

- Typical Table Size-median and mode is 2 this confirm customer behaviour is driven by couples or pairs
- Small Groups-mean is higher than mode indicating pair aee common but there are enough group of 3 or 4 people to nudge average up.

EXECUTIVE SUMMARY FOR STRATEGY

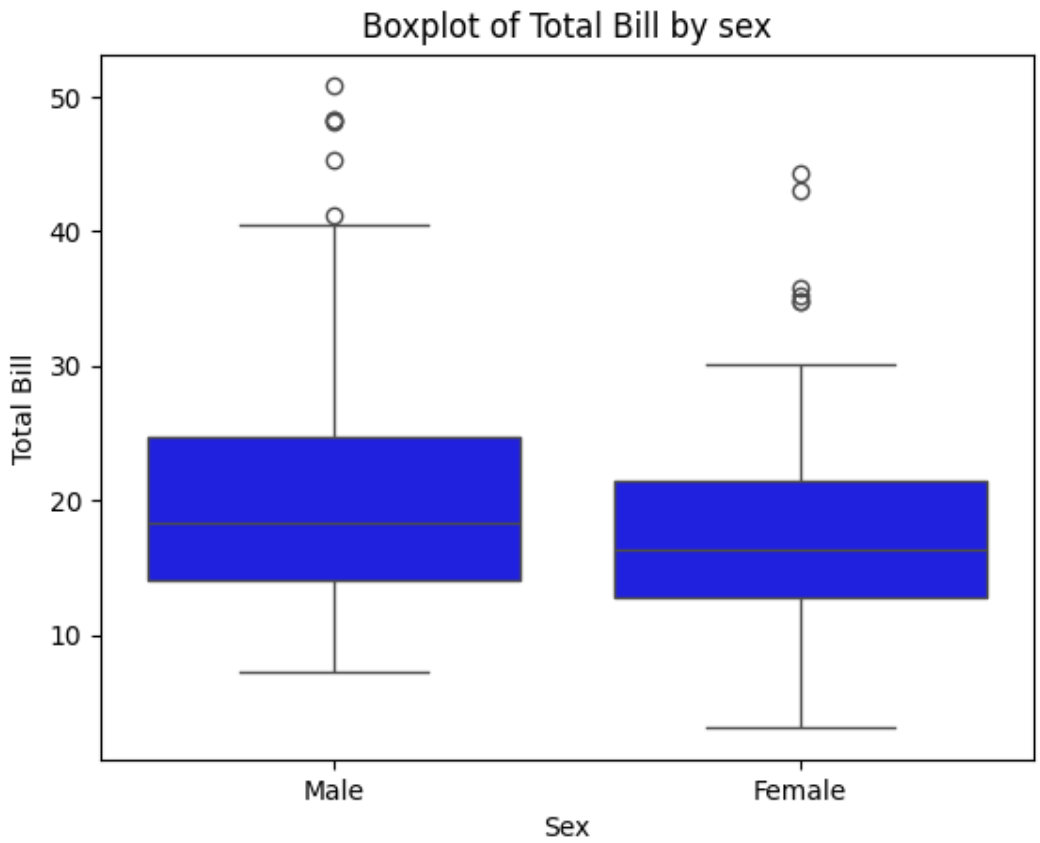
Metric	Customer Behaviour Insight
Primary target	Pairs/couples(mode size-2)
Price sensitivity	High(most frequent bill is 13.42 well below the ,mean)
Revenue driver	Occasional high spend table (indicated by 47.74 range)
Service value	Tips are consistent suggesting steady service satisfaction.



VISUAL TRENDS FROM HISTORGAM OF TOTAL_BILL

HIGHEST BAR	15-18$	MOST COMMON BILL
OVERALL SHAPE	Right Skewed	Asymmetry (Most Bill Are Cheaper And Very Less Bill Are Higherside)
DENSITY	10-25$ Bar Is High	70-80% Customer Bill Are Around 25 Dollar
RIGHT TALE	Between 40-50	Big Spender Or They Are Outlier

VISUAL TRENDS:BOXPLOT(BILL)



KEY COMPONENT	MALE	FEMALE	COMPARISON INSIGHT
Median	*18-20$*	*16-17$*	*On average male is paying more*
IQR	*Large box*	*Small box*	*Female spending more consistent because of small box show less variance than male with large box.*
Whisker	*7-41$(min-max)*	*3-30$(min-max)*	*High bill of male is more than female*
Outliers	*4-5 circle with spending gap*	*3-4 circle but overlapping less gap in extreme spending*	*Extreme spending in both group but in male group it is more extreme.*

CONCLUSION

- Men generally spend more and have more variety in their bills compared to women while both group mostly spend between 13$ and 25$.
- Average spending on weeken is highest.

- **Correlation analysis**

Total bill vs size-0.59	It Means As Number Of People Increase Bill Goes Up As It Is Not Closer To One It Means There Are Other Factor Like Two People Also Order Sometime More.
Total bill vs tip-0.67	Strength Between Tip And Total Bill Is Strong It Means Higher Sales Leads To Higher Tips